

DAOD 3026-0, Radio Frequency Safety

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1. Introduction

Date of Issue: 2012-05-04

Date of Last Modification: 2019-02-05

Application: This DAOD is a directive that applies to employees of the Department of National Defence (DND employees) and an order that applies to officers and non-commissioned members of the Canadian Armed Forces (CAF members).

Supersession: CFAO 34-51, Radio Frequency Radiation Safety

Approval Authority: Assistant Deputy Minister (Materiel) (ADM(Mat))

Enquiries: Quality Engineering Test Establishment (QETE) Superintendent

2. Definitions

ammunition (munition) - A device charged with explosives, propellants, pyrotechnics, initiating composition or nuclear, biological or chemical material, for use in military operations, and includes a non-charged or inert replica of such a device. (Defence Terminology Bank record number 15923)

emission (émission) - The phenomenon by which energy emanates from a source in the form of waves or particles. (Defence Terminology Bank record number 7989)

explosives (explosif) - Any thing that is made, manufactured or used to produce an explosion or a detonation or pyrotechnic effect, and includes any thing prescribed to be an explosive by the regulations made under the Explosives Act, but does not include gases, organic peroxides or any thing prescribed not to be an explosive by the regulations made under that Act. (Section 2 of the Explosives Act)

exposure (exposition) - The amount of absorbed radiation as determined by the level of emission and the time duration of that emission. (Defence Terminology Bank record number 48285)

radio frequency (radiofréquence) - The frequency of a periodic radio wave or of the corresponding electrical oscillation. (Defence Terminology Bank record number 7982)

radio frequency device (dispositif de radiofréquence) - A device that generates or utilizes radio frequency energy. (Defence Terminology Bank record number 48268)

radio frequency radiation (rayonnement de radiofréquence) - Energy propagating through space in form of electromagnetic waves, which: are comprised of electric and magnetic fields oscillating at frequencies in the range of 3 kHz to 300 GHz; and do not contain enough energy to remove tightly bound electrons from their orbits around atoms. (Defence Terminology Bank record number 48269)

secondary x-ray (rayon X secondaire) - Unintended X-ray radiation emitted from a radio frequency device containing an internal high voltage source exceeding 15 kilovolts. (Defence Terminology Bank record number 693843)

3. Policy direction

Context

3.1 This DAOD deals with all radio frequency (RF) devices owned or operated by the DND and the CAF that: operate in the frequency range from 3 kHz to 300 GHz; and are used during training and operations, including research and development activities.

Policy Statement

3.2 The DND and the CAF are committed to: preventing incidents, injuries and losses that may occur as a result of RF radiation exposures; and planning, implementing and managing activities in the DND and the CAF that create RF radiation within a safety management framework, consistent with applicable practices and standards, to the extent practicable in a military context.

Requirements

3.3 The DND and the CAF must: ensure DND employees, CAF members and members of the public are not exposed to harmful levels of RF radiation and secondary X-rays; develop and implement a Radio Frequency Safety Program (RFSP) that: ensures all DND employees, CAF members and contractor personnel at defence establishments, including any vehicle, ship or aircraft, during training and operations, minimize their RF radiation exposures and that such exposures are within the limits set out in the Limits of Human Exposure to Radiofrequency Electromagnetic Energy in the Frequency Range from 3 kHz to 300 GHz – Safety Code 6 (2015) of Health Canada; provides guidance to prevent or control hazards associated with RF devices; includes a process to evaluate and address hazards associated with RF devices; and ensures RFSP requirements are taken into account during the materiel acquisition and support process; and ensures the appropriate level of training of DND employees, CAF members and other persons granted access to defence establishments while working with or near RF devices; and implement a compliance evaluation process to ensure that: RF devices meet RFSP requirements; and appropriate physical and administrative hazard controls for RF devices are established.

4. Consequences

General

4.1 Should clarification of the policy statement, requirements or authorities set out in this DAOD be required, DND employees and CAF members may seek direction through their channel of communication or chain of command, as appropriate. The anticipated results set out in the policy statement may not be achieved by the DND and the CAF if the requirements specified in this DAOD are not properly implemented. Not achieving the anticipated results could affect the ability of the DND and the CAF to ensure that the CAF is prepared to undertake missions for the protection of Canada and Canadians and the maintenance of international peace and stability.

5. Authorities

Authority Table

5.1 The following table identifies the authorities associated with this DAOD:

ADM(Mat) - issue policies, orders, directives and guidance for the RFSP; and establish the framework for the Radio Frequency Safety Orders and Directives.

L1 advisors whose organizations hold or use RF devices - issue, within the framework established by the RFSP, RF safety (RFS) orders, directives and guidance respecting the use of RF devices in their organizations.

Director General Land Equipment Program Management - act as the RFSP authority and RFSP Champion; ensure RFSP integrity; and provide guidance and oversight for all DND and CAF RF activities.

Director Ammunition and Explosives Regulation - issue policies for DND and CAF activities resulting in RF radiation hazards to ammunition and explosives; establish compliance monitoring of RF radiation hazards to ammunition and explosives; and oversee DND and CAF activities related to RF radiation hazards to ammunition and explosives.

QETE Superintendent - oversee RFSP operations and integrity; establish requirements and procedures for the RFSP; establish RFS standards for RF radiation hazards to personnel, ammunition, explosives, fuel and other materiel; appoint the RFSP manager and the RFS technical authority; and represent the DND and the CAF on matters related to RFS, jointly with the RFSP manager, in dealings with Health Canada, Innovation, Science and Economic Development Canada, other government departments, North Atlantic Treaty Organization (NATO) allies and international organizations.

Director Nuclear Safety - issues policies, orders, directives and guidance for secondary X-ray monitoring, evaluation and controls.

managers of a DND organization, commanders of a command or formation, and COs of a unit that hold or use RF devices - appoint RFS officers.

RFSP manager - oversee: RFSP policy development and program management; RFS compliance monitoring; and all DND and CAF RF activities conducted in accordance with DAOD 3026-1, Radio Frequency Safety Program; and represent the DND and the CAF on matters related to RFS, jointly with the QETE Superintendent, in dealings with Health Canada, Innovation, Science and Economic Development Canada, other government departments, NATO allies and international organizations.

chairperson of the Radio Frequency Safety Board - advise L1 organizations and operational commands on: risks associated with the use of RF devices, including RF radiation hazards to personnel, fuel, ammunition, explosives and other materiel; measures to prevent injury and control RF radiation exposures; and RF devices and their proposed uses prior to acquisition or deployment.

6. References

Acts, Regulations, Central Agency Policies and Policy DAOD:

Explosives Act

Canada Labour Code, Part II, Occupational Health and Safety

Canada Occupational Health and Safety Regulations

Limits of Human Exposure to Radiofrequency Electromagnetic Energy in the Frequency Range from 3 kHz to 300 GHz - Safety Code 6 (2015), Health Canada

DAOD 1000-8, Policy Framework for Safety and Security Management

Other References:

DAOD 2007-0, Safety

DAOD 3000-0, Materiel Acquisition and Support

DAOD 3002-0, Ammunition and Explosives

DAOD 3002-3, Ammunition and Explosives Safety Program

DAOD 3026-1, Radio Frequency Safety Program

DAOD 4002-1, Nuclear and Ionizing Radiation Safety

A-GG-040-001/AG-001, General Safety Program, Volume 1, Policy and Program

A-GG-040-006/AG-001, DND Explosives Safety Program

C-09-005-009/TS-001, Ammunition and Explosives Safety Manual, Volume 9, Hazards of Electromagnetic Radiation to Ordnance (HERO)

C-55-040-001/TS-002, Radio Frequency Safety Standards and Requirements

CF H Svcs Gp Instruction 4440-16, Physical Hazards Surveillance Program

Radio Standards Specification 102, Radio Frequency (RF) Exposure Compliance of Radiocommunication Apparatus (All Frequency Bands), Innovation, Science and Economic Development Canada

GL-01, Guidelines for the Measurement of Radio Frequency Fields at Frequencies from 3 KHz to 300 GHz, Innovation, Science and Economic Development Canada

STANAG 2345, Military Workplaces – Force Health Protection Regarding Personnel Exposure to Electric, Magnetic, and Electromagnetic Fields, 0 Hz to 300 GHz